Year	Quarter	Sales	GNP	d1 =IF($B2=1,1,0)	d2 =IF($B2=2,1,0)	d3 =IF($B2=3,1,0)
2011	1	1059989	15285.828	1	0	0
2011	2	1161474	15496.189	0	1	0
2011	3	1167991	15591.850	0	0	1
2011	4	1222523	15796.460	0	0	0
2012	1	1141734	16019.758	1	0	0
2012	2	1212532	16152.257	0	1	0
2012	3	1214923	16257.151	0	0	1
2012	4	1272567	16358.863	0	0	0
2013	1	1171293	16569.591	1	0	0
2013	2	1260446	16637.926	0	1	0
2013	3	1272129	16848.748	0	0	1
2013	4	1316396	17083.137	0	0	0
2014	1	1195724	17104.555	1	0	0
2014	2	1323909	17432.909	0	1	0
2014	3	1327929	17721.657	0	0	1
2014	4	1379952	17849.912	0	0	0
2015	1	1232060	18003.399	1	0	0
2015	2	1354216	18223.577	0	1	0
2015	3	1367232	18347.425	0	0	1
2015	4	1412991	18378.803	0	0	0
2016	1	1278391	18470.156	1	0	0
2016	2	1386918	18656.207	0	1	0
2016	3	1399285	18821.359	0	0	1
2016	4	1461780	19032.580	0	0	0
2017	1	1324253	19237.435	1	0	0
2017	2	1445499	19379.232	0	1	0
2017	3	1451087	19617.288	0	0	1
2017	4	1536250	19937.963	0	0	0
2018	1	1390044	20242.215	1	0	0
2018	2	1519069	20552.653	0	1	0
2018	3	1536768	20742.723	0	0	1
2018	4	1585439	20909.853	0	0	0
2019	1	1417491	21115.309	1	0	0
2019	2	1574816	21329.877	0	1	0
2019	3	1595623	21540.325	0	0	1
2019	4	1652271	21747.394	0	0	0
2020	1	1443243	21561.139	1	0	0
2020	2	1455588	19520.114	0	1	0
2020	3	1639409	21170.252	0	0	1
2020	4	1718793	21487.896	0	0	0

a. Estimate the model where y and x represent retail sales and GNP, respectively; d1 equals 1 if quarter 1, 0 otherwise; d2 equals 1 if quarter 2, 0 otherwise; and d3 equals 1 if quarter 3, 0 otherwise. Note that the reference category is quarter 4.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.989573594
R Square	0.979255898
Adjusted R Square	0.976885144
Standard Error	23982.156
Observations	40

อธิบายความแปรปรวนของ Earnings เป็น 97.92% (ควรมากกว่า70%)

ANOVA	degree of freedom	SS	SS/DF MS	MSR/MST F	Significance F
	df	*SS*	*MS*	*F*	*Significance F*
R Regression	4	9.50268E+11	2.37567E+11	413.0566518	6.3712E-29
E Residual	35	20130033216	575143806.2		
T Total	39	9.70398E+11			

		Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
b0	Intercept	135609.2001	37106.93833	3.654551041	0.000837066	60278.11042	210940.2898	60278.11042	210940.2898
b1	GNP	70.01097517	1.92613954	36.34782098	2.1772E-29	66.10070402	73.92124632	66.10070402	73.92124632
b2	d1	-155654.2095	10767.84334	-14.4554675	2.52107E-16	-177514.0937	-133794.3254	-177514.0937	-133794.3254
b3	d2	-50030.3508	10771.8471	-4.644547064	4.67157E-05	-71898.36299	-28162.33861	-71898.36299	-28162.33861
b4	d3	-45187.90729	10731.54739	-4.210754109	0.000168966	-66974.10673	-23401.70784	-66974.10673	-23401.70784

RESIDUAL OUTPUT

Observation	*Predicted Sales*	*Residuals*		**Y = b0*1+b1*x1+b2*x2+b3*x3+b4*x5+b5*x5+b6*x6**
1	1050130.715	9858.284857	1050130.715	=$J$20*1+$J$21*D2+$J$22*E2+$J$23*F2+$J$24*G2
2	1170482.153	-9008.15261	1170482.153	=$J$20*1+$J$21*D3+$J$22*E3+$J$23*F3+$J$24*G3
3	1182021.916	-14030.916	1182021.916	=$J$20*1+$J$21*D4+$J$22*E4+$J$23*F4+$J$24*G4
4	1241534.769	-19011.7689	1241534.769	=$J$20*1+$J$21*D5+$J$22*E5+$J$23*F5+$J$24*G5
5	1101513.87	40220.12985	1101513.87	=$J$20*1+$J$21*D6+$J$22*E6+$J$23*F6+$J$24*G6
6	1216414.113	-3882.11307	1216414.113	=$J$20*1+$J$21*D7+$J$22*E7+$J$23*F7+$J$24*G7
7	1228600.288	-13677.2878	1228600.288	=$J$20*1+$J$21*D8+$J$22*E8+$J$23*F8+$J$24*G8
8	1280909.151	-8342.15141	1280909.151	=$J$20*1+$J$21*D9+$J$22*E9+$J$23*F9+$J$24*G9
9	1140008.215	31284.78534	1140008.215	=$J$20*1+$J$21*D10+$J$22*E10+$J$23*F10+$J$24*G10
10	1250416.273	10029.72663	1250416.273	=$J$20*1+$J$21*D11+$J$22*E11+$J$23*F11+$J$24*G11
11	1270018.571	2110.429307	1270018.571	=$J$20*1+$J$21*D12+$J$22*E12+$J$23*F12+$J$24*G12
12	1331616.28	-15220.2804	1331616.28	=$J$20*1+$J$21*D13+$J$22*E13+$J$23*F13+$J$24*G13
13	1177461.566	18262.43402	1177461.566	=$J$20*1+$J$21*D14+$J$22*E14+$J$23*F14+$J$24*G14
14	1306073.808	17835.19156	1306073.808	=$J$20*1+$J$21*D15+$J$22*E15+$J$23*F15+$J$24*G15
15	1331131.781	-3202.78102	1331131.781	=$J$20*1+$J$21*D16+$J$22*E16+$J$23*F16+$J$24*G16
16	1385298.946	-5346.94592	1385298.946	=$J$20*1+$J$21*D17+$J$22*E17+$J$23*F17+$J$24*G17
17	1240390.511	-8330.51094	1240390.511	=$J$20*1+$J$21*D18+$J$22*E18+$J$23*F18+$J$24*G18
18	1361429.246	-7213.24616	1361429.246	=$J$20*1+$J$21*D19+$J$22*E19+$J$23*F19+$J$24*G19
19	1374942.409	-7710.40893	1374942.409	=$J$20*1+$J$21*D20+$J$22*E20+$J$23*F20+$J$24*G20
20	1422327.121	-9336.12059	1422327.121	=$J$20*1+$J$21*D21+$J$22*E21+$J$23*F21+$J$24*G21
21	1273068.624	5322.376319	1273068.624	=$J$20*1+$J$21*D22+$J$22*E22+$J$23*F22+$J$24*G22
22	1391718.094	-4800.09434	1391718.094	=$J$20*1+$J$21*D23+$J$22*E23+$J$23*F23+$J$24*G23
23	1408122.99	-8837.99043	1408122.99	=$J$20*1+$J$21*D24+$J$22*E24+$J$23*F24+$J$24*G24
24	1468098.686	-6318.68591	1468098.686	=$J$20*1+$J$21*D25+$J$22*E25+$J$23*F25+$J$24*G25
25	1326786.575	-2533.5747	1326786.575	=$J$20*1+$J$21*D26+$J$22*E26+$J$23*F26+$J$24*G26
26	1442337.78	3161.220336	1442337.78	=$J$20*1+$J$21*D27+$J$22*E27+$J$23*F27+$J$24*G27
27	1463846.756	-12759.7559	1463846.756	=$J$20*1+$J$21*D28+$J$22*E28+$J$23*F28+$J$24*G28
28	1531485.433	4764.567364	1531485.433	=$J$20*1+$J$21*D29+$J$22*E29+$J$23*F29+$J$24*G29
29	1397132.202	-7088.20233	1397132.202	=$J$20*1+$J$21*D30+$J$22*E30+$J$23*F30+$J$24*G30
30	1524490.128	-5421.12816	1524490.128	=$J$20*1+$J$21*D31+$J$22*E31+$J$23*F31+$J$24*G31
31	1542639.558	-5871.55773	1542639.558	=$J$20*1+$J$21*D32+$J$22*E32+$J$23*F32+$J$24*G32
32	1599528.399	-14089.3993	1599528.399	=$J$20*1+$J$21*D33+$J$22*E33+$J$23*F33+$J$24*G33
33	1458258.365	-40767.3647	1458258.365	=$J$20*1+$J$21*D34+$J$22*E34+$J$23*F34+$J$24*G34
34	1578904.338	-4088.33832	1578904.338	=$J$20*1+$J$21*D35+$J$22*E35+$J$23*F35+$J$24*G35
35	1598480.452	-2857.45154	1598480.452	=$J$20*1+$J$21*D36+$J$22*E36+$J$23*F36+$J$24*G36
36	1658165.461	-5894.46145	1658165.461	=$J$20*1+$J$21*D37+$J$22*E37+$J$23*F37+$J$24*G37
37	1489471.358	-46228.3577	1489471.358	=$J$20*1+$J$21*D38+$J$22*E38+$J$23*F38+$J$24*G38
38	1452201.066	3386.934132	1452201.066	=$J$20*1+$J$21*D39+$J$22*E39+$J$23*F39+$J$24*G39
39	1572571.28	66837.72007	1572571.28	=$J$20*1+$J$21*D40+$J$22*E40+$J$23*F40+$J$24*G40
40	1639997.753	78795.24659	1639997.753	=$J$20*1+$J$21*D41+$J$22*E41+$J$23*F41+$J$24*G41

b. Interpret the slope coefficient for quarter 1.

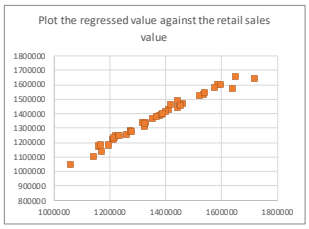
Coefficients quarter 1 คือ −155,654.21 หมายความว่ายอดขายในไตรมาสที่ 1 จะ ต่ำกว่าไตรมาสที่ 4 ซึ่งเป็นกลุ่มอ้างอิง อยู่ประมาณ 155,654.21 ล้านดอลลาร์

c. What are predicted sales in quarter 2 if GNP is $21,000 (in billions)? For the same value of GNP, what are predicted sales in quarter 4?

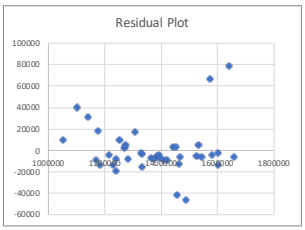
quarter 2	1,555,809.33	millions	=($J$20*1)+($J$21*21000)+($J$23*1)	d2 = 1
quarter 4	1,605,839.68	millions	=($J$20*1)+($J$21*21000)	d4 = ไม่มี ก็ไม่ต้องบวก

d. Plot (1) the regressed value against the retail sales value and (2) residual, (3) do your think linear regression is fitted for this problem, why or why not?

(1)Plot the regressed value against the retail sales value



(2) residual



(3) do your think linear regression is fitted for this problem, why or why not?
โมเดลถดถอยเชิงเส้น (Linear Regression) เป็นโมเดลที่เหมาะสมกับปัญหานี้
- เนื่องจาก R Square = 0.9793 หรือ 97.93%ซึ่งหมายความว่า โมเดลสามารถอธิบายความแปรปรวนของยอดขายได้มากถึง 97.93% ถือว่าโมเดลมีความสามารถในการพยากรณ์ที่แม่นยำและครอบคลุมข้อมูลส่วนใหญ่

- จากกราฟ "Plot the regressed value against the retail sales value"จะเห็นว่า จุดข้อมูลส่วนใหญ่อยู่ใกล้กับเส้นแนวโน้มที่เป็นเส้นตรงแสดงให้เห็นว่ามีความสัมพันธ์เชิงเส้นที่ชัดเจนระหว่างค่าทำนายและค่าจริงซึ่งเป็นคุณสมบัติสำคัญของโมเดล Linear Regression ที่ดี

- จากกราฟ Residual Plot ซึ่งแสดงค่าความคลาดเคลื่อน (Residual)พบว่าจุดกระจายแบบสุ่มรอบแกนศูนย์ โดยไม่มีรูปแบบชัดเจน เช่น โค้ง ฟุ่ง หรือกระจุก

โจทย์

A government researcher is analyzing the relationship between retail sales (Sales in $ millions) and the gross national product (GNP in $ billions). He also wonders whether there are significant differences in retail sales related to the quarters of the year. He collects 10 years of quarterly data and creates dummy variables for the quarters. A portion is shown in Table 7.5.

TABLE 7.5 Data for Example 7.4

Year	Quarter	Sales	GNP
2011	1	1059989	15285.828
2011	2	1161474	15496.189
⋮	⋮	⋮	⋮
2020	4	1718793	21487.896

a. Estimate the model $y = \beta_0 + \beta_1 x + \beta_2 d_1 + \beta_3 d_2 + \beta_4 d_3 + \varepsilon$, where y and x represent retail sales and GNP, respectively; d1 equals 1 if quarter 1, 0 otherwise; d2 equals 1 if quarter 2, 0 otherwise; and d3 equals 1 if quarter 3, 0 otherwise. Note that the reference category is quarter 4.
b. Interpret the slope coefficient for quarter 1.
c. What are predicted sales in quarter 2 if GNP is $21,000 (in billions)? For the same value of GNP, what are predicted sales in quarter 4?
d. Plot (1) the regressed value against the retail sales value and (2) residual, (3) do your think linear regression is fitted for this problem, why or why not?